

Linear Initial-Value Problem for the Finite-Depth Rayleigh–Taylor Instability

Abstract

This note summarizes the linear initial-value formulation implemented in the latest Mathematica notebook. Two inviscid, incompressible, irrotational fluids are separated by a nominally flat interface at $z = 0$ and are bounded by rigid horizontal walls at $z = H_1$ and $z = -H_2$. The initial interface is flat, but an interfacial vertical perturbation velocity $v_0(x)$ is imposed at $t = 0$. The resulting dynamics are solved mode by mode using the finite-depth Rayleigh–Taylor dispersion relation with surface tension.

1 Geometry and assumptions

The upper fluid occupies

$$0 < z < H_1, \tag{1}$$

while the lower fluid occupies

$$-H_2 < z < 0. \tag{2}$$

The unperturbed interface is located at $z = 0$. The densities are denoted by ρ_1 for the upper fluid and ρ_2 for the lower fluid. The gravitational acceleration is g , and the surface tension coefficient is σ .

The interface displacement is denoted by

$$z = \eta(x, t), \tag{3}$$

where η is assumed sufficiently small for linearization. The rigid-wall impermeability conditions are

$$w_1(x, H_1, t) = 0, \quad w_2(x, -H_2, t) = 0, \tag{4}$$

where w_1 and w_2 are the vertical velocities in the upper and lower fluids, respectively.

The initial-value problem considered in the notebook is

$$\eta(x, 0) = 0, \quad \partial_t \eta(x, 0) = v_0(x). \tag{5}$$

Thus, the interface is initially flat, and the perturbation is introduced through the initial vertical velocity at the interface.

2 Mathematical background

This section records the mathematical ingredients used later in the note. The goal is to make clear where the modal equation, the finite-depth factors, and the verification tests come from.

2.1 Linearized potential-flow model

Each fluid is assumed inviscid, incompressible, and irrotational. Therefore the perturbation velocity in fluid j can be represented by a velocity potential

$$\mathbf{u}_j = (u_j, w_j) = \nabla \phi_j, \quad j = 1, 2. \quad (6)$$

Incompressibility gives

$$\nabla \cdot \mathbf{u}_j = 0, \quad (7)$$

and hence the potentials satisfy Laplace's equation in their respective fluid domains,

$$\partial_{xx}\phi_j + \partial_{zz}\phi_j = 0. \quad (8)$$

The rigid walls are impermeable, so the normal velocity vanishes there:

$$\partial_z \phi_1(x, H_1, t) = 0, \quad \partial_z \phi_2(x, -H_2, t) = 0. \quad (9)$$

The interface is the material surface $F(x, z, t) = z - \eta(x, t) = 0$. The kinematic condition $D_t F = 0$ gives

$$\partial_t \eta + u_j \partial_x \eta = w_j \quad \text{on } z = \eta(x, t). \quad (10)$$

After linearizing about the flat state $\eta = 0$ and neglecting products of perturbation quantities, this becomes

$$\partial_t \eta(x, t) = \partial_z \phi_1(x, 0, t) = \partial_z \phi_2(x, 0, t). \quad (11)$$

This condition is why the interface velocity used in the notebook is simply $\partial_t \eta$.

2.2 Fourier modes and finite-depth eigenfunctions

Because the coefficients and boundaries are independent of x , a Fourier mode evolves independently in the linear theory. For a nonzero mode

$$\phi_j(x, z, t) = \hat{\phi}_j(z, t) e^{ikx}, \quad k > 0, \quad (12)$$

Laplace's equation (8) reduces to

$$\frac{d^2 \hat{\phi}_j}{dz^2} - k^2 \hat{\phi}_j = 0. \quad (13)$$

The finite-depth wall conditions (9) select hyperbolic functions. Writing the modal interfacial velocity as

$$\widehat{W}(t) = \partial_t \widehat{\eta}(t), \quad (14)$$

the vertical velocities are

$$\widehat{w}_1(z, t) = \widehat{W}(t) \frac{\sinh(k(H_1 - z))}{\sinh(kH_1)}, \quad (15)$$

$$\widehat{w}_2(z, t) = \widehat{W}(t) \frac{\sinh(k(z + H_2))}{\sinh(kH_2)}. \quad (16)$$

These shapes equal $\widehat{W}$ at the interface and vanish at the rigid walls. Integrating once in z gives the interface values of the velocity potentials,

$$\hat{\phi}_1(0, t) = -\frac{\coth(kH_1)}{k} \partial_t \widehat{\eta}(t), \quad \hat{\phi}_2(0, t) = \frac{\coth(kH_2)}{k} \partial_t \widehat{\eta}(t). \quad (17)$$

The factors $\rho_1 \coth(kH_1)$ and $\rho_2 \coth(kH_2)$ that appear in the dispersion relation come from these interface potential values. They represent the finite-depth inertial loading of the two fluids on the moving interface.

2.3 Dynamic condition and surface tension

The linearized unsteady Bernoulli relation gives the perturbation pressure in each fluid. Evaluated at the displaced interface and retaining the hydrostatic contribution from moving the interface by η , the pressure jump can be written as

$$p_1 - p_2 = -\rho_1 \partial_t \phi_1(x, 0, t) + \rho_2 \partial_t \phi_2(x, 0, t) - (\rho_1 - \rho_2)g\eta. \quad (18)$$

With the sign convention used in this note, the linearized capillary pressure jump is

$$p_1 - p_2 = \sigma \partial_{xx} \eta. \quad (19)$$

Equivalently,

$$\rho_1 \partial_t \phi_1(x, 0, t) - \rho_2 \partial_t \phi_2(x, 0, t) + (\rho_1 - \rho_2)g\eta = -\sigma \partial_{xx} \eta. \quad (20)$$

For a Fourier mode, $-\partial_{xx} \eta = k^2 \eta$. Substituting (17) into (20) gives

$$-\frac{\rho_1 \coth(kH_1) + \rho_2 \coth(kH_2)}{k} \partial_{tt} \hat{\eta} + (\rho_1 - \rho_2)g\hat{\eta} = \sigma k^2 \hat{\eta}. \quad (21)$$

Solving for $\partial_{tt} \hat{\eta}$ yields the modal growth coefficient used throughout the note.

2.4 Stability information in the dispersion relation

For $k > 0$ the denominator

$$D(k) = \rho_1 \coth(kH_1) + \rho_2 \coth(kH_2) \quad (22)$$

is positive. Hence stability is determined by the sign of the numerator

$$k(\rho_1 - \rho_2)g - \sigma k^3. \quad (23)$$

The gravitational term is destabilizing when the upper fluid is heavier than the lower fluid, $\rho_1 > \rho_2$. The capillary term is stabilizing and grows like k^3 , so sufficiently short waves oscillate rather than grow.

The finite-depth factors modify the effective inertia. In the deep-water limit $kH_1, kH_2 \gg 1$, $\coth(kH_j) \rightarrow 1$ and the denominator approaches $\rho_1 + \rho_2$. In the shallow, long-wave limit $kH_j \ll 1$, $\coth(kH_j) \sim 1/(kH_j)$, so the denominator becomes large and the growth of very long waves is reduced by the rigid boundaries.

2.5 Initial-value interpretation

The modal equation has the form

$$\frac{d^2 \hat{\eta}}{dt^2} = \lambda \hat{\eta}, \quad (24)$$

where $\lambda = \omega^2(k)$ in the convention of this note. The kernel $G(\lambda, t)$ used below is the Green's function for the initial data

$$G(\lambda, 0) = 0, \quad \partial_t G(\lambda, 0) = 1. \quad (25)$$

Therefore $\hat{\eta}(t) = \hat{v} G(\lambda, t)$ automatically enforces a flat initial interface and an imposed initial velocity $\hat{v}$. Positive λ gives hyperbolic growth, negative λ gives trigonometric oscillation, and the case $\lambda = 0$ is obtained by the continuous limit $G(0, t) = t$.

For a real initial velocity profile, the Fourier coefficients satisfy the conjugate symmetry

$$\hat{v}_{-n} = \overline{\hat{v}_n}. \quad (26)$$

The reconstructed interface is real for this reason; taking the real part in the implementation is a robust numerical way to remove tiny imaginary round-off residuals.

2.6 Discrete projection and verification

The notebook computes Fourier coefficients with a discrete periodic projection. This is the discrete Fourier transform normalized to match the continuous convention

$$\hat{v}_n = \frac{1}{L} \int_0^L v_0(x) e^{-ik_n x} dx. \quad (27)$$

For smooth periodic data, the periodic trapezoidal rule converges rapidly, and for trigonometric data whose modes are resolved by the grid it recovers the exact coefficients up to floating-point round-off.

The verification tests follow directly from the kernel and shape functions. Since $G(\lambda, 0) = 0$, one should have $\eta(x, 0) = 0$. Since $\partial_t G(\lambda, 0) = 1$, one should have $\partial_t \eta(x, 0) = v_0(x)$ after reconstruction. Finally, the hyperbolic sine factors in (15)–(16) vanish at $z = H_1$ and $z = -H_2$, so the wall velocity checks should be zero up to numerical round-off.

3 Finite-depth dispersion relation

For a Fourier mode with horizontal wavenumber $k > 0$, the corrected finite-depth dispersion relation used in the notebook is

$$\omega^2(k) = \frac{k(\rho_1 - \rho_2)g - \sigma k^3}{\rho_1 \coth(kH_1) + \rho_2 \coth(kH_2)}. \quad (28)$$

The sign of the capillary term is stabilizing: for sufficiently large k , the term $-\sigma k^3$ can make $\omega^2(k)$ negative, producing stable oscillatory modes rather than exponential growth.

When $\rho_1 > \rho_2$, the critical wavenumber at which the numerator changes sign is

$$k_c = \sqrt{\frac{(\rho_1 - \rho_2)g}{\sigma}}. \quad (29)$$

For $0 < k < k_c$, the numerator of (28) is positive, and the corresponding modes are Rayleigh–Taylor unstable. For $k > k_c$, surface tension stabilizes the mode in the linear theory.

4 Modal initial-value problem

Consider a periodic domain of length L . The horizontal wavenumbers are

$$k_n = \frac{2\pi n}{L}, \quad n \in \mathbb{Z}. \quad (30)$$

The zero mode is omitted in the notebook, since the instability calculation concerns nonzero interfacial perturbations. The initial velocity is expanded as

$$v_0(x) = \sum_{n \neq 0} \hat{v}_n e^{ik_n x}, \quad (31)$$

with Fourier coefficients

$$\hat{v}_n = \frac{1}{L} \int_0^L v_0(x) e^{-ik_n x} dx. \quad (32)$$

For each nonzero mode, define

$$\lambda_n = \omega^2(|k_n|). \quad (33)$$

The modal interface amplitude satisfies

$$\frac{d^2 \hat{\eta}_n}{dt^2} = \lambda_n \hat{\eta}_n, \quad (34)$$

with initial conditions

$$\hat{\eta}_n(0) = 0, \quad \frac{d\hat{\eta}_n}{dt}(0) = \hat{v}_n. \quad (35)$$

The solution is

$$\hat{\eta}_n(t) = \hat{v}_n G(\lambda_n, t), \quad (36)$$

where

$$G(\lambda, t) = \begin{cases} \frac{\sinh(\sqrt{\lambda} t)}{\sqrt{\lambda}}, & \lambda \neq 0, \\ t, & \lambda = 0. \end{cases} \quad (37)$$

Its time derivative is

$$\partial_t G(\lambda, t) = \begin{cases} \cosh(\sqrt{\lambda} t), & \lambda \neq 0, \\ 1, & \lambda = 0. \end{cases} \quad (38)$$

If $\lambda < 0$, these formulae are interpreted through complex arguments. Equivalently, writing $\lambda = -\alpha^2$ with $\alpha > 0$, one obtains

$$G(-\alpha^2, t) = \frac{\sin(\alpha t)}{\alpha}, \quad \partial_t G(-\alpha^2, t) = \cos(\alpha t), \quad (39)$$

which corresponds to stable capillary-gravity oscillations.

The reconstructed interface displacement is

$$\eta(x, t) = \text{Re} \left[\sum_{n \neq 0} \hat{\eta}_n(t) e^{ik_n x} \right]. \quad (40)$$

The interfacial vertical velocity is

$$w(x, 0, t) = \partial_t \eta(x, t) = \text{Re} \left[\sum_{n \neq 0} \hat{v}_n \partial_t G(\lambda_n, t) e^{ik_n x} \right]. \quad (41)$$

5 Vertical velocity fields

The finite-depth vertical structure used in the notebook satisfies the impermeability conditions at the two walls. If the interfacial vertical velocity amplitude of mode n is denoted by

$$\hat{w}_n(0, t) = \hat{v}_n \partial_t G(\lambda_n, t), \quad (42)$$

then the upper-fluid vertical velocity is reconstructed using

$$w_{1,n}(z, t) = \hat{w}_n(0, t) \frac{\sinh(|k_n|(H_1 - z))}{\sinh(|k_n|H_1)}. \quad (43)$$

Similarly, the lower-fluid vertical velocity is

$$w_{2,n}(z, t) = \hat{w}_n(0, t) \frac{\sinh(|k_n|(z + H_2))}{\sinh(|k_n|H_2)}. \quad (44)$$

Therefore,

$$w_1(x, z, t) = \text{Re} \left[\sum_{n \neq 0} \hat{w}_n(0, t) \frac{\sinh(|k_n|(H_1 - z))}{\sinh(|k_n|H_1)} e^{ik_n x} \right], \quad (45)$$

for $0 < z < H_1$, and

$$w_2(x, z, t) = \operatorname{Re} \left[\sum_{n \neq 0} \hat{w}_n(0, t) \frac{\sinh(|k_n|(z + H_2))}{\sinh(|k_n|H_2)} e^{ik_n x} \right], \quad (46)$$

for $-H_2 < z < 0$.

These formulae immediately imply

$$w_1(x, H_1, t) = 0, \quad w_2(x, -H_2, t) = 0, \quad (47)$$

and

$$w_1(x, 0, t) = w_2(x, 0, t) = \partial_t \eta(x, t). \quad (48)$$

6 Example used in the notebook

The notebook uses the parameter set

$$\rho_1 = 2, \quad \rho_2 = 1, \quad g = 9.81, \quad \sigma = 0.01, \quad H_1 = H_2 = 1, \quad L = 2\pi. \quad (49)$$

The imposed initial interfacial velocity is

$$v_0(x) = U_0 \cos(3x) + \frac{2}{5} U_0 \sin(5x), \quad U_0 = 10^{-3}. \quad (50)$$

Since $L = 2\pi$, the nonzero exact Fourier coefficients are

$$\hat{v}_3 = \frac{U_0}{2}, \quad \hat{v}_{-3} = \frac{U_0}{2}, \quad (51)$$

$$\hat{v}_5 = -\frac{iU_0}{5}, \quad \hat{v}_{-5} = \frac{iU_0}{5}. \quad (52)$$

The numerical notebook nevertheless computes the coefficients through a discrete periodic projection, which is more general and can be used for arbitrary smooth periodic profiles.

7 Discrete Fourier projection in the Mathematica notebook

The notebook uses a uniform grid

$$x_m = \frac{mL}{N}, \quad m = 0, 1, \dots, N-1, \quad (53)$$

where $N = 4096$. The Fourier coefficient is approximated by the periodic trapezoidal rule

$$\hat{v}_n \approx \frac{1}{N} \sum_{m=0}^{N-1} v_0(x_m) e^{-ik_n x_m}. \quad (54)$$

This avoids the convergence warnings caused by numerical quadrature when the exact coefficient is zero. The Mathematica implementation also applies `Chop` to remove round-off residuals.

8 Verification conditions

The notebook verifies the initial conditions by computing

$$\max_x |\eta(x, 0)|, \quad \max_x |\partial_t \eta(x, 0) - v_0(x)|. \quad (55)$$

For a sufficiently resolved Fourier truncation, both quantities should be close to machine precision.

The wall boundary conditions are verified by computing

$$\max_x |w_1(x, H_1, 0)|, \quad \max_x |w_2(x, -H_2, 0)|. \quad (56)$$

Because the vertical shape functions vanish exactly at the corresponding rigid walls, these errors should also be at the level of numerical round-off.

A Core Mathematica implementation

The following listing contains the essential part of the corrected Mathematica notebook.

```
Remove["Global`*"];

pars = {
  rho1 -> 2.0,
  rho2 -> 1.0,
  g -> 9.81,
  sigma -> 0.01,
  H1 -> 1.0,
  H2 -> 1.0,
  L -> 2 Pi
};

Lnum = N[L /. pars];

omega2[k_] := (k (rho1 - rho2) g - sigma k^3)/
  (rho1 Coth[k H1] + rho2 Coth[k H2]);

omega2Num[k_?NumericQ] := N[omega2[k] /. pars];

kCritical = Sqrt[((rho1 - rho2) g)/sigma] /. pars;

tol = 10^-12;

growthKernel[lambda_?NumericQ, t_?NumericQ] :=
  If[Abs[lambda] < tol,
    t,
    Sinh[Sqrt[lambda] t]/Sqrt[lambda]
  ];

growthKernelDot[lambda_?NumericQ, t_?NumericQ] :=
  If[Abs[lambda] < tol,
    1,
    Cosh[Sqrt[lambda] t]
  ];

U0 = 10^-3;

v0Expr = U0 Cos[3 x] + (2/5) U0 Sin[5 x];
```

```

v0[xVal_?NumericQ] := N[v0Expr /. x -> xVal];

nMax = 32;

kn[n_Integer] := 2 Pi n/Lnum;

modes = Join[Range[-nMax, -1], Range[1, nMax]];

nSamples = 4096;

xGrid = N[Lnum Range[0, nSamples - 1]/nSamples];
v0Grid = v0 /@ xGrid;

Clear[vHat];

vHat[n_Integer] := vHat[n] =
  Chop[
    (1/nSamples) Total[v0Grid Exp[-I kn[n] xGrid]],
    10^-14
  ];

etaHat[n_Integer, tVal_?NumericQ] :=
  vHat[n] growthKernel[omega2Num[Abs[kn[n]]], tVal];

eta[xVal_?NumericQ, tVal_?NumericQ] :=
  Re[
    Total[
      etaHat[#, tVal] Exp[I kn[#] xVal] & /@ modes
    ]
  ];

wInterfaceHat[n_Integer, tVal_?NumericQ] :=
  vHat[n] growthKernelDot[omega2Num[Abs[kn[n]]], tVal];

wInterface[xVal_?NumericQ, tVal_?NumericQ] :=
  Re[
    Total[
      wInterfaceHat[#, tVal] Exp[I kn[#] xVal] & /@ modes
    ]
  ];

shape1[k_?NumericQ, zVal_?NumericQ] :=
  Sinh[k ((H1 /. pars) - zVal)]/Sinh[k (H1 /. pars)];

shape2[k_?NumericQ, zVal_?NumericQ] :=
  Sinh[k (zVal + (H2 /. pars))]/Sinh[k (H2 /. pars)];

w1[xVal_?NumericQ, zVal_?NumericQ, tVal_?NumericQ] :=
  Re[
    Total[
      (wInterfaceHat[#, tVal]
       shape1[Abs[kn[#]], zVal]
       Exp[I kn[#] xVal]) & /@ modes
    ]
  ];

w2[xVal_?NumericQ, zVal_?NumericQ, tVal_?NumericQ] :=

```



```

Re[
  Total[
    (wInterfaceHat[#, tVal]
     shape2[Abs[kn[#]], zVal]
     Exp[I kn[#] xVal]) & /@ modes
  ]
];

checkGrid = N[Lnum Range[0, 512]/512];

maxEtaInitialError =
  Max[Abs[eta[#, 0] & /@ checkGrid]];

maxVelocityInitialError =
  Max[Abs[(wInterface[#, 0] - v0[#]) & /@ checkGrid]];

maxUpperWallError =
  Max[Abs[w1[#, H1 /. pars, 0] & /@ checkGrid]];

maxLowerWallError =
  Max[Abs[w2[#, -(H2 /. pars), 0] & /@ checkGrid]];

{
  maxEtaInitialError,
  maxVelocityInitialError,
  maxUpperWallError,
  maxLowerWallError
}

```